Idemia Datachallenge (IADATA704): Face Occlusion Prediction

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Project repository: https://github.com/odelacruz-fierro/data-challenge-idemia.git

1 Introduction

In the context of the Data Challenge proposed by Idemia, the objective is to predict the percentage of facial occlusion in images (224x224 pixels). The major challenge lies in handling the training data imbalance (a low proportion of the training database contain highly occluded faces) and the gender fairness requirement imposed by the evaluation metric.

In this report, we present our solution based on multi-task learning and data augmentaion, designed to extract robust features while minimizing the demographic bias penalty.

2 Exploratory Data Analysis

The exploratory data analysis (EDA) performed on the train database revealed that only a low proportion of the faces show a high occlusion. This is verified by Fig.1, where we can observe a right-skewed distribution: the samples are more concentrated on the lower values than in the extreme high values, whici drag the tail towards the right side.

A closer look to different occlusion levels (see Table 1) allows us to notice that the majority of images concentrate in the occlusion interval $[0.0, 0.5]$ with a minority of them in $(0.5, 1.0]$. In order to understand what thes intervals looked-like, four occlusion levels were proposed: **low** $\in [0.0, 0.2]$, **medium-low** $\in (0.2, 0.4]$, **medium-high** $\in (0.4, 0.5]$ and **high** $\in (0.5, 1.0]$. Sample images for these intervals are provided in figure 2.

- **low** occluded images trend to provide a global visibility of the face forehead, eyes, nose, chicks and chin.
- **medium-low** and **medium-high** occluded images par of the faces are covered either with the hair, a hat, hand, fingers and even physical objects.
- Finally in **high** occluded images faces seem to be mostly visible, however a blurry / noised effect was noticed.

From a gender perspective, it was found that the train database contains 67.6 % of men and 32.4 % of women (see Fig.3).

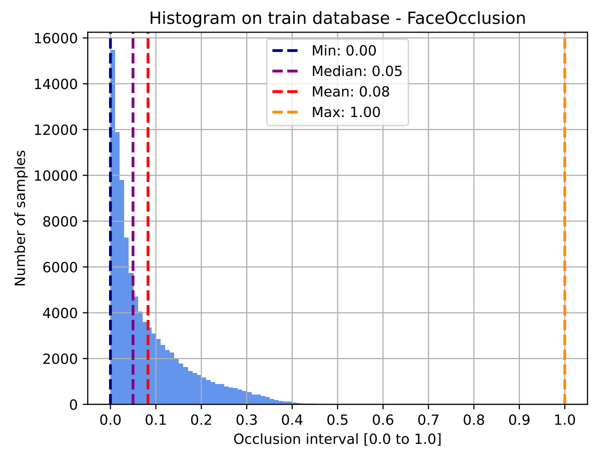

Figure 1: Distribution of occlusion across the training database.

3 Methodology & Architecture

TO DO ...

3.1 Milti-Task Learning (MLT)

TO DO ...

3.2 Backbone architectures

TO DO ...

4 Training strategy

TO DO ...

4.1 Custom loss

TO DO ...

4.2 Data augmentation

Often real-world data sets are predominately composed of “normal” examples with only a small percentage of “abnormal” or “interesting” examples. The goal of this section is to increase the diversity of the train database in order to match close enough the distribution of the test database (see Fig.4).

Visual analysis of the sampled intervals (see section 2) revealed a distinct dichotomy in the training data: true structural occlusions (e.g., hands, helmets, instruments, and hair) dominate the $[0.20, 0.50]$ range, whereas values above 0.50 consist primarily of quality degradations

Table 1: Distribution of images across occlusion intervals in the training database

Occlusion interval	**Image Count**	**(%)**
[0.00, 0.10]	68,929	68.93%
(0.10, 0.20]	19,558	19.56%
(0.20, 0.30]	8,372	8.37%
(0.30, 0.40]	2,865	2.86%
(0.40, 0.50]	240	0.24%
(0.50, 0.60]	26	0.03%
(0.60, 0.70]	5	0.01%
(0.70, 0.80]	2	0.00%
(0.80, 0.90]	0	0.00%
(0.90, 1.00]	3	0.00%
Total	**100,000**	**100.00%**

such as severe sensor pixelation and artistic filter artifacts. Given that the target test distribution concentrates heavily below 0.50, we prioritize `RandomErasing` as our primary spatial regularizer to simulate realistic object boundaries, treating `GaussianBlur` as a secondary augmentation to handle baseline quality invariance.

Reference test [1]
Reference test [2]
Reference test [3]

5 Results

TO DO ...

6 Limitations and tradeoffs

7 Conclusion

TO DO ...

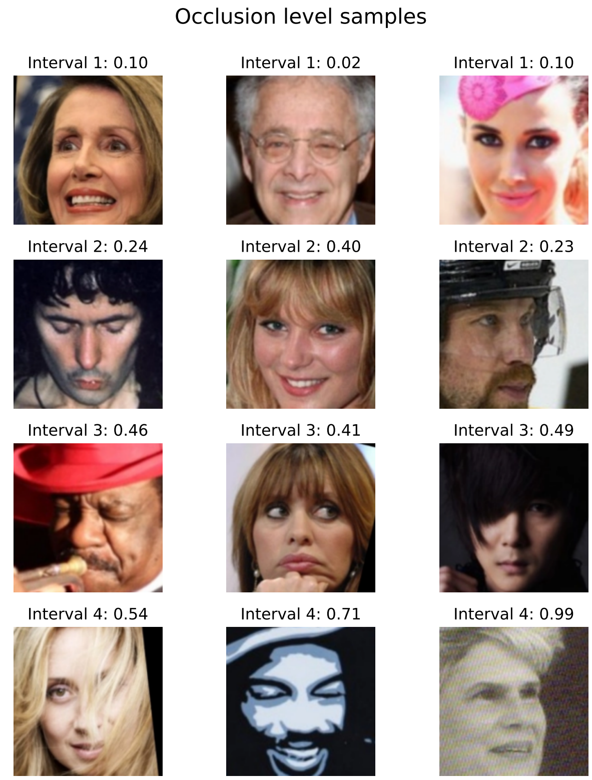

Figure 2: Random samples for four occlusion intervals.

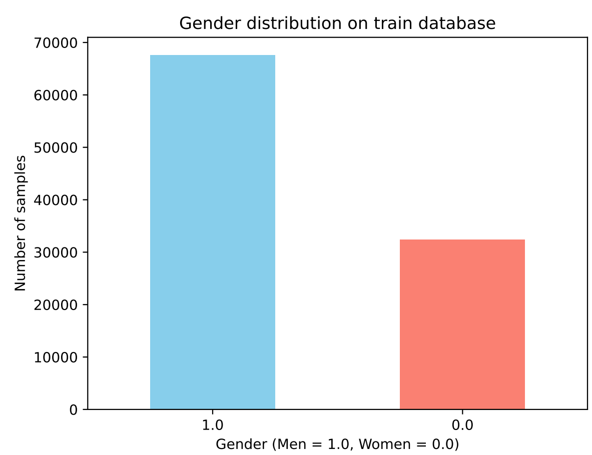

Figure 3: Distribution of gender across the training dataset: Men = 67.6 % and Women = 32.4 %.

References

[1] Mingle Xu, Sook Yoon, Alvaro Fuentes, and Dong Sun Park. A Comprehensive Survey of Image Augmentation Techniques for Deep Learning. *Pattern Recognition*, 137:109347, May 2023.

[2] Pengguang Chen, Shu Liu, Hengshuang Zhao, Xingquan Wang, and Jiaya Jia. GridMask Data Augmentation, January 2020.

[3] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer. SMOTE: Synthetic Minority Over-sampling Technique. *Journal of Artificial Intelligence Research*, 16:321–357, June 2002.

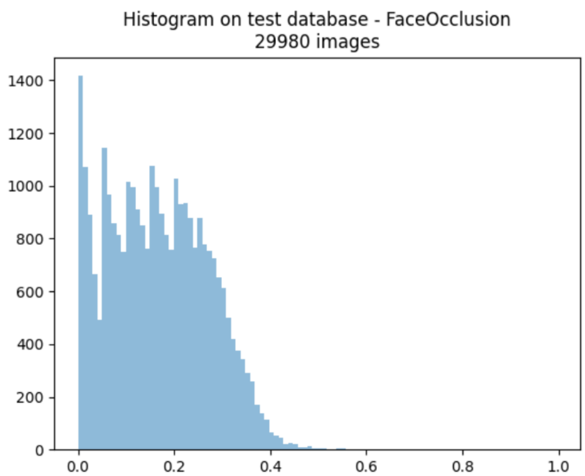

Figure 4: Distribution of occlusion across the test database.